Adarsh Kumar Thakur

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adarsh-fpga.github.io/MyPortfolio

Skills

Technical: SystemVerilog, Verilog, VHDL, UVM, Advanced Digital Design, C, C++, Communication Protocol, Assertion & Code Coverage, Static Timing Analysis, Computer Architecture

Tools: Vivado, Vitis, QuestaSim, ActiveHDL, LTspice, ModelSim

Work Experience

CDAC Chennai | R&D **Apr 2024 – Present**

Project Engineer

Design and Verification of biometric enabled Indigenous FIDO2 Authenticator

- Designed and implemented a hardware crypto engine integrating SHA-256, HMAC, AES (GCM/ECB/CBC/CTR), and ECDSA modules in Verilog.
- Implemented PQC algorithms (ML-KEM, ML-DSA) in the crypto engines.
- Integrated the crypto core with FPGA and SoC platforms for secure authentication flow.
- Performed FPGA synthesis, STA, and on-board validation.
- Packaged RTL as Vivado IP and integrated AXI, UART, GPIO, timers, and memory.
- Integrated the core with a RISC-V SoC and executed HW–SW co-verification.
- Integrated GT521F32 fingerprint sensor to enable biometric based authentication.
- Verified AXI/AXI4-Stream using constrained-random testing and achieved coverage closure.
- Built UVM environment with SVA, scoreboards, and functional coverage.
- Developed and deployed the RP and FIDO2 server, including a web interface for authenticator testing.
- Performed system-level validation and hardware testing for FIDO-based secure operations.

CDAC Hyderabad **Sep 2023 – Feb 2024**

VLSI Trainee

- Specialized in Digital Design, Computer Architecture, SoC design, C, and Python with hands-on experience in Verilog, SystemVerilog, and FPGA tools.
- Performed end-to-end AXI4-Stream verification by developing a SystemVerilog-based testbench.

Education

Chandigarh University, University Institute of Engineering **June 2023**

B.E. in Electronics and Communication Engineering **CGPA: 7.11/10**

Relevant Coursework: VLSI Design, Digital System Design, Computer Architecture, Low Power VLSI, Verilog HDL, Signal and System, Microprocessors and Microcontrollers

Chhotu Ram Rural Institute of Technology, Delhi **June 2020**

Diploma in Electronics and Communication Engineering **CGPA: 7.00/10**

Project Work

- **UVM-Based AXI4-Stream VIP (Jan 2024 – Feb 2024):** Developed a configurable AXI4-Stream VIP supporting both master and slave modes. Built a UVM-based SystemVerilog testbench with assertions and functional coverage. Verified protocol compliance using constrained-random testing and coverage analysis.
- **Gas Sensor System (Feb 2023 – June 2023):** Designed a gas detection system with high sensitivity and selectivity. Implemented signal processing and alarm indication features. Achieved reliable detection performance with optimized response time.
- **Head-Tilt Controlled Wheelchair (Aug 2022 – Dec 2022):** Developed a control system using head-motion sensing for wheelchair navigation. Translated head tilt angles into directional movement commands. Integrated safety features including speed control and emergency stop.

Publications

- **PQC-FIDO: Post-Quantum FIDO2 Authentication with ML-DSA on Constrained Dual-Core Hardware. A. K. Thakur** (First author). *The 29th International Conference on Advanced Communications Technology (ICACT 2027)*. Under review.
- **Th-FIDO: Dynamic-Quorum Registration and Zero-Touch Recovery for Threshold FIDO2 Authentication. A. K. Thakur**, et al. (First author). *The 22nd ACM ASIA Conference on Computer and Communications Security (ACM ASIACCS 2027)*, Macau, July 2027. Preprint.

Courses and Certificates

- PG - Diploma in VLSI from CDAC - Hyderabad (R&D)
- TIMTS and AICRA (Robotics); NPTEL (Sensors and Actuators)
- Secure Processor 2026 workshop by SETS & CDAC; Digital Systems: from Logic Gates to Processors